Data Pipeline in Practice

Marco Morales
marco.morales@columbia.edu

Nana Yaw Essuman
ne2388@columbia.edu

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for Social Scientists

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recap: workflow principles in the Data Science Shop

a) **reproducibility**
   - anyone should be able to arrive to your **same results**

b) **portability**
   - anyone should be able to **pick up where you left off** on any machine

crucial tenets for **collaborative work**

a) **scalability**
   - your project should also work for **larger data sets** and/or be on the path of **automation**

recap: structuring your workspace

```
workspace
|
| -- /src
|      |-- /data              <- code to read/munge raw data
|      |-- /features          <- code to transform/append data
|      |-- /models            <- code to analyze data
|      |-- /visualizations    <- code to create visualizations
|      |-- /functions         <- scripts to centralize functions
|      |-- /config            <- configuration files
|
| -- README.md                <- high-level project description
```

Pro tips

- data is NEVER pushed to GitHub!!!!!!
- {secret keys} are NEVER pushed to GitHub!!!!!!
- references are transferred to GitHub wiki
- TODO is transferred to GitHub projects

question:

what is data?

definition

for our purposes, **data** is

information (as opposed to pure instructions)

encoded in a digital (binary) format

recorded and stored in an **electronic** form

question:

why is <u>data collection</u> important?

- understand your **products and systems** better
- provide means for organizations to make **better data-informed decisions**
- help identify **opportunities or gaps** in a product or system
- measure how your **consumers interact** with your products or system
- understand your **potential market**

In God we trust, all others bring data.

–William E. Deming

but…not all **data** is created equal!

- **Unstructured data**
  - does not have a predefined data model or is not organized in a pre-defined manner
  - examples of unstructured data include audio, video files or No-SQL databases.

- **Structured data**
  - pre-defined data model and ready to analyze
  - examples of structured data are Excel files or SQL databases
  - most traditional form of data storage

but…not all **data** is created equal!

structured

Family Name	Given Name	VIAF ID
Ackersdijck	Willem Cornelis	17959345
Adelung	Friedrich von	22963658
Afzelius	Arvid August	49972119
Amerling	Karel	13331054

```
Family Name,Given Name,VIAF ID
Ackersdijck,Willem Cornelis,17959345
Adelung,Friedrich von,22963658
Afzelius,Arvid August,49972119
Amerling,Karel,13331054
```

.csv .TSV Parquet

- fixed structure (schema)

semi-structured

```
{
  "customerID": "103248",
  "name":
  {
    "first": "AAA",
    "last": "BBB"
  },
  "address":
  {
    "street": "Main Street",
    "number": "101",
    "city": "Acity",
    "state": "NY"
  },
  "ccOnFile": "yes",
  "firstOrder": "02/28/2003"
}
```

```
<?xml version="1.0" encoding="UTF-8"?>
<events>
    <event>
        <week>-mtwtf-</week>
        <starttime>06:00</starttime>
        <endtime>11:59</endtime>
        <playlist>morning.m3u8</playlist>
    </event>
    <event>
        <week>SMTWTFS</week>
        <starttime>12:00</starttime>
        <endtime>20:59</endtime>
        <playlist>afternoon.m3u8</playlist>
    </event>
</events>
```

{JSON} <xml />

- no fixed structure (schema)
- some metadata (*e.g.* tags)

unstructured

text image audio video LOG logs

- no pre-defined structure (schema)

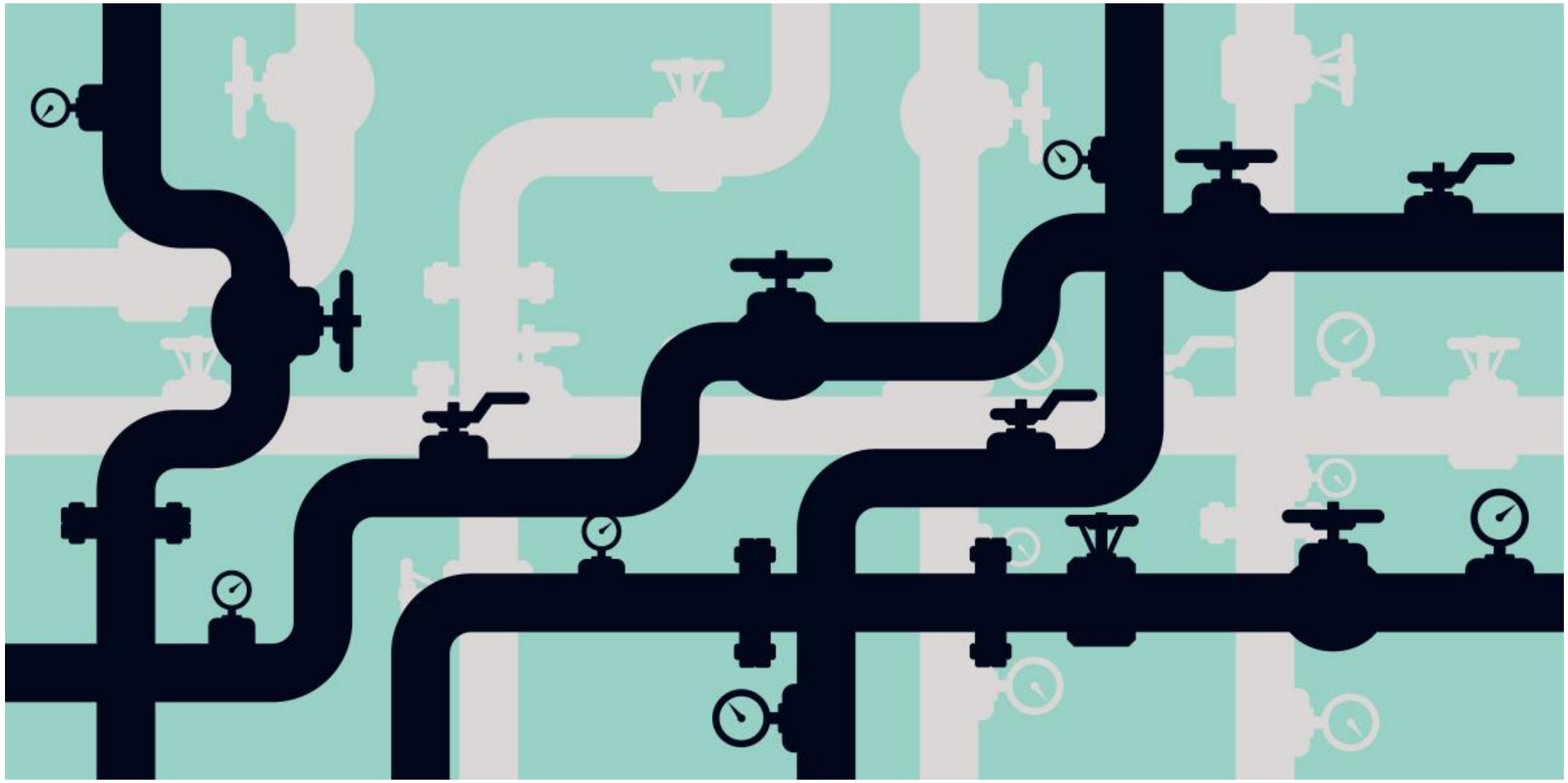
levels of datasets

- **First party datasets**
  - data generated by your **own product or systems**
  - the most useful and valuable data you can collect about **your consumers**
- **Second party datasets**
  - **someone else's first-party data** but useful to your organization
  - arrangement with trusted partners who are willing to share their customer data with you (and vice versa)
- **Third party datasets**
  - data that is **widely accessible to competitors**, so you aren't gaining unique advantage
  - great for demographic, behavioral, and contextual targeting
  - data that you buy from outside sources that are not the original collectors of that data (**data aggregators**)

question:

how do we get data?

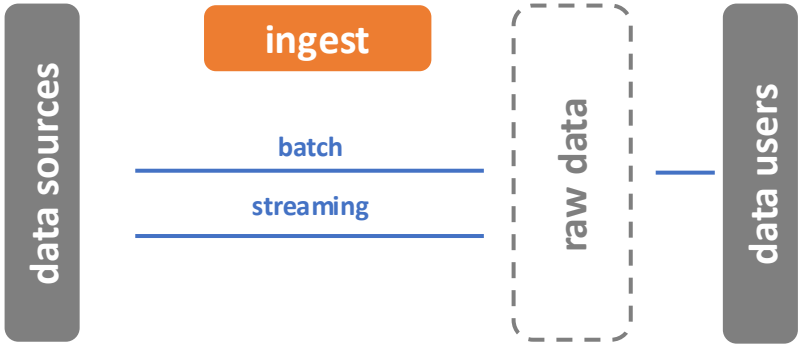
let’s introduce a concept: **data pipelines**



a **process** to **move data** across systems

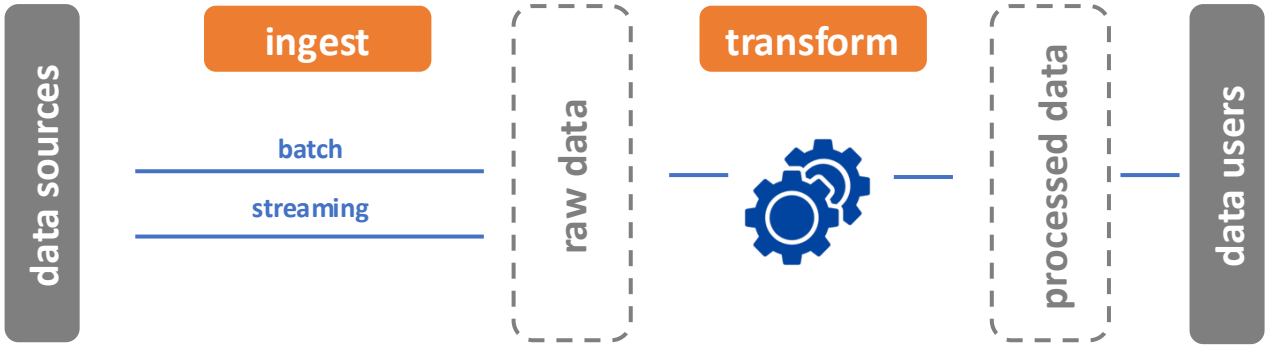
a data pipeline

- "**pull**" data in
- "**push**" data to new location

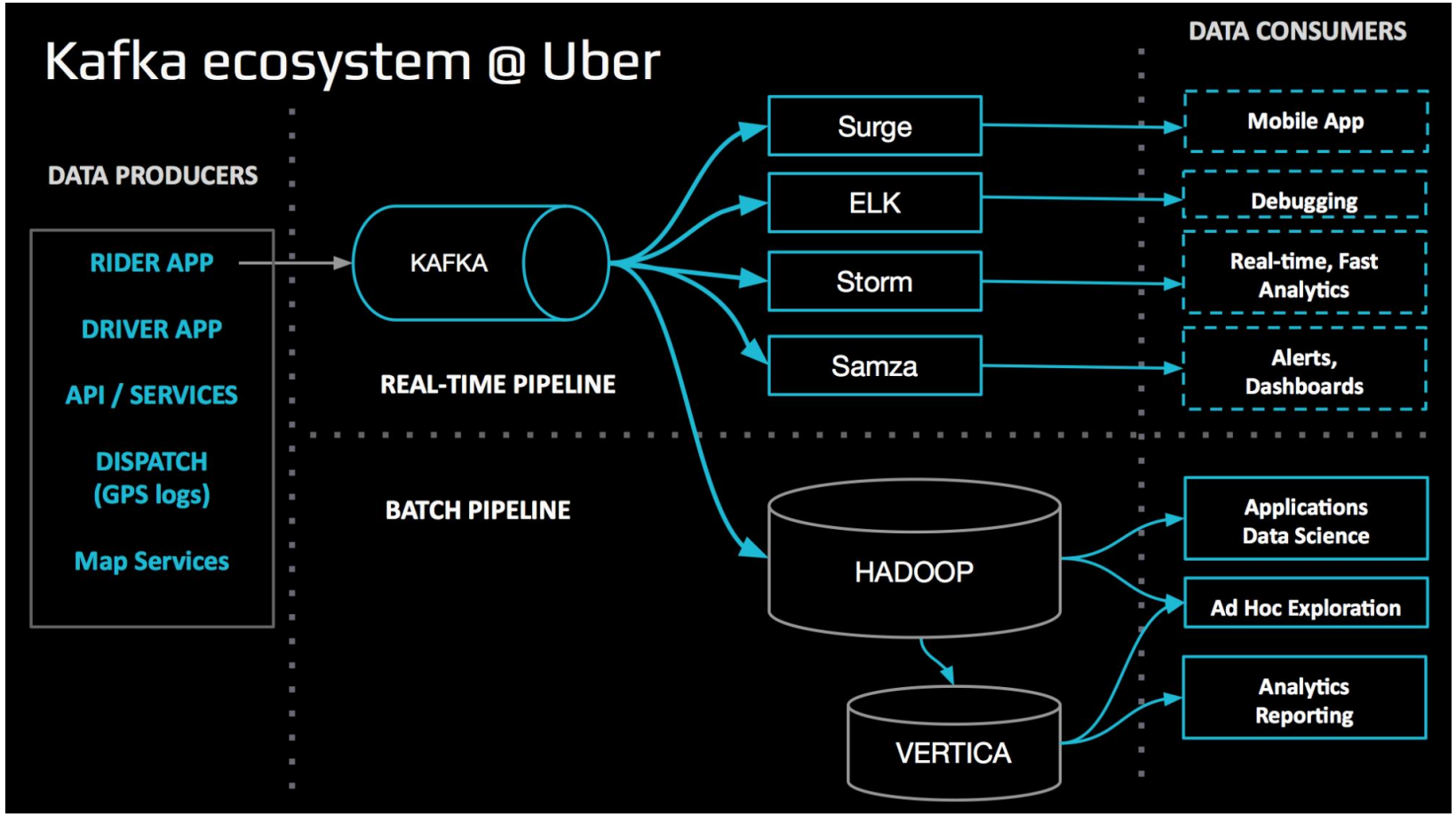


also a data pipeline

- "**pull**" data in
- **transform** data
- "**push**" data to new location



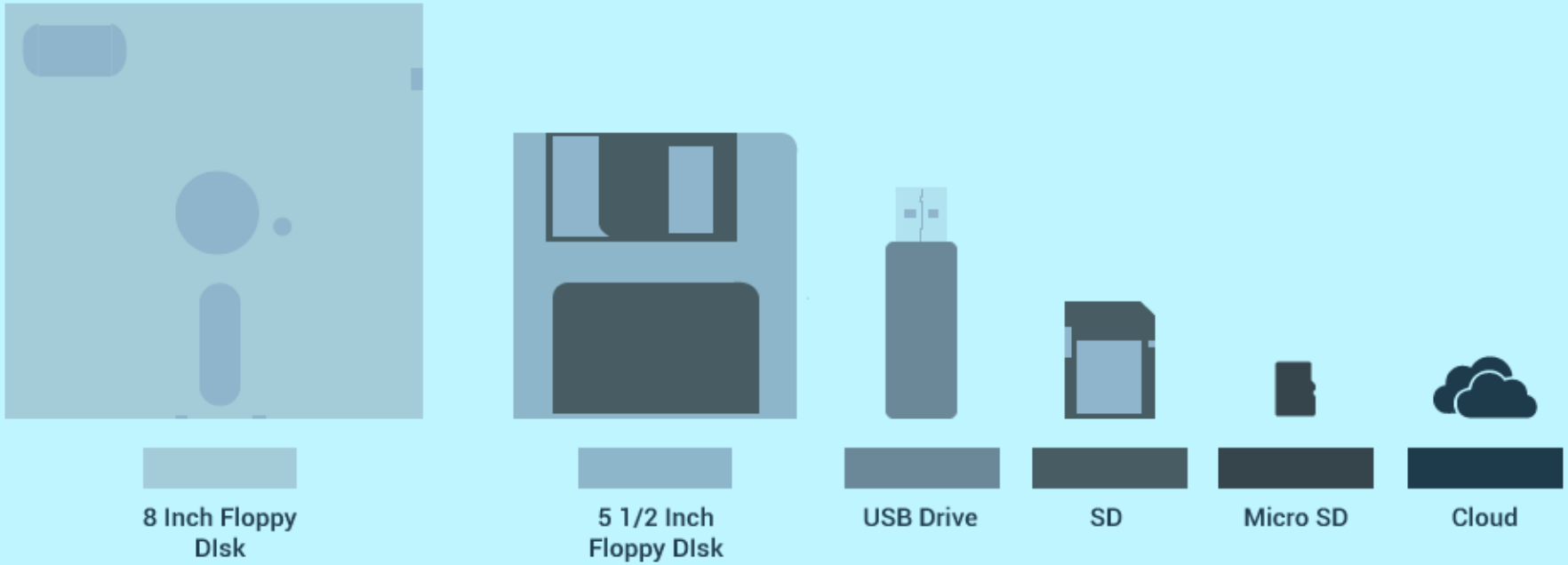
a **data pipeline** matches its **use case**



question:

how is data stored?

The Evolution of Data Storage



	8 Inch Floppy DIsk	5 1/2 Inch Floppy DIsk	USB Drive	SD	Micro SD	Cloud
Commercially available	1971	1976	2000	2010	2010	2009
Maximum capacity	1.2 mb	1.2 mb	2 tb	256 gb	128 gb	~5zb
Cost per gb	£1000	£800	£0.5	£0.8	£0.5	<50gb free

@vowlesyy @pixelpelican pixelpelican.co.uk samuelvowles.co.uk
sources: Wikipedia, Amazon, Ebay

Ways of storing data

- **Object storage**
  - is a way of structuring stored data so that it’s characterized as objects that can be manipulated in different ways by hardware and network storage systems
  - the objects are not in a file-folder hierarchy
  - object stores are scalable, fast data retrieval and cost effective

- **Distributed file systems**
  - a file system with data stored on a server.
  - data is accessed and processed as if it was stored on the local client machine
  - convenient to share information and files among users on a network in a controlled and authorized way

Ways of storing data

- **Relational Databases**
  - uses a structure that allows us to identify and access data in relation to another piece of data in the database
  - data in a relational database is organized into tables

- **NoSQL Databases**
  - a non-relational way of storing data
  - mostly used to store documents, key-value pair data
  - storing a large volume of data, and you don’t want to lock yourself into a schema

Hands on workshop

question:

how is data managed?

data management for each age…

data lake

- **raw** data (central) repository
- **large** volumes
- multiple **sources/formats**

structured | unstructured | semi-structured

circa 2010s

Google Cloud Storage
Microsoft Azure Data Lake
Amazon S3

data warehouse

- **processed** data (central) repository
- **smaller** volumes
- **structured & normalized**

structured | semi-structured

circa 1990s

ORACLE
snowflake
teradata.

data lakehouse

- **data processing** on top of **raw data**
- **on-demand** (streaming)
- supports **multiple** use cases

structured | unstructured | semi-structured

circa 2020s

Google Cloud
databricks
DELTA LAKE
ORACLE Cloud Infrastructure
aws

perhaps more intuitively…

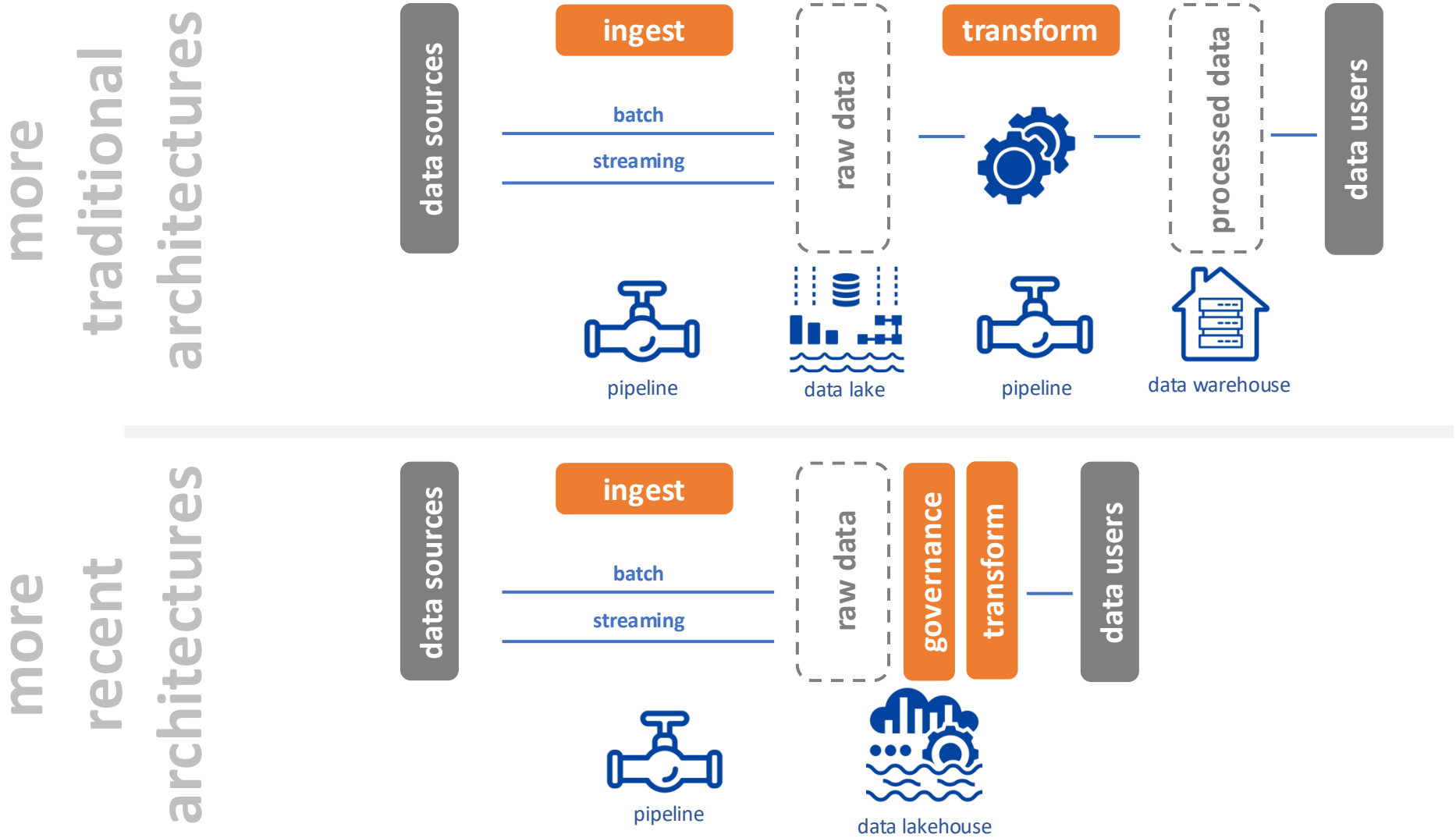
perhaps more intuitively…

a data lake

a data warehouse

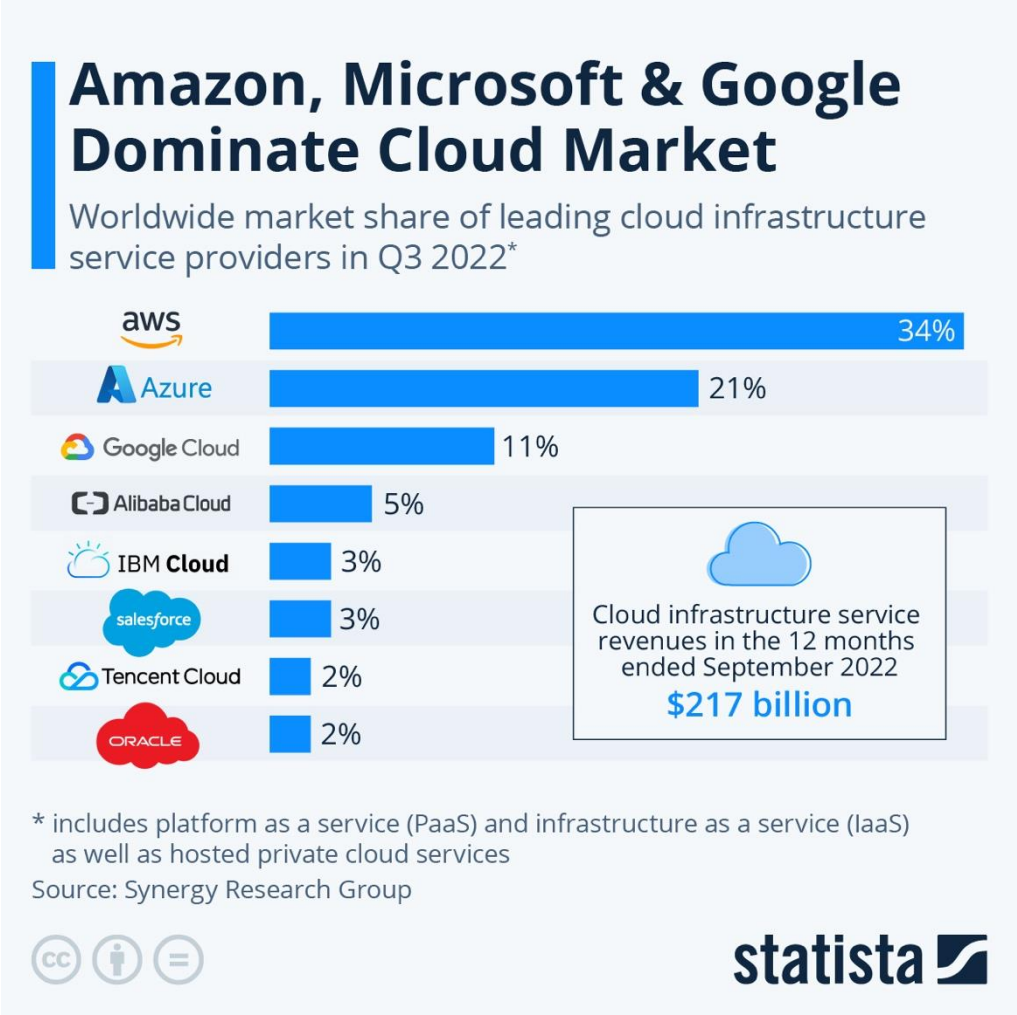
a data lakehouse

data architectures to bring it all together



question:
who provides these services?

just a **cloud** away!



tools for this class

getting started with Apache Spark

- an **open-source distributed cluster-computing** framework
- provides an interface for programming entire clusters with **implicit data parallelism** and **fault tolerance**
- has **5 main components**:
  - Spark Core
  - Spark SQL
  - Spark Streaming
  - MLlib
  - GraphX
- Databricks was formed by the creators of Apache Spark to make it more efficient to utilize Spark and manage Spark Clusters

setting up Databricks

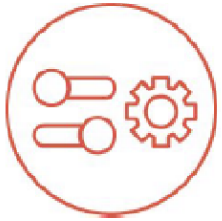
databricks®

getting started with cloud computing

- on-demand availability of computer system resources, especially data storage and computing power, without direct active management by the user
- ability to scale elastically - for example, more or less computing power, storage, bandwidth-right when they're needed, and from the right geographic location
- three main deployment models of Cloud Computing:
  - Private Cloud
  - Public Cloud
  - Hybrid cloud

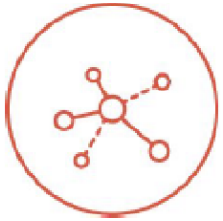
cloud computing in a nutshell

What is cloud computing?



On-demand self-service

No human intervention needed to get resources



Broad network access

Access from anywhere

Resource pooling

Provider shares resources to customers

Rapid elasticity

Get more resources quickly as needed

Measured service

Pay only for what you consume

setting up Amazon Web Services (AWS)

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